

Coulomb's Law

- Coulomb's law states that the electrical force between two charged objects is directly proportional to the product of the quantity of charge on the objects and inversely proportional to the square of the separation distance between the two objects.

$$F = k \frac{q_1 q_2}{r^2}$$

Where q_1 and q_2 are amount of charge on two objects, r is the distance between the charge bodies and k is the coulomb's constant having value of $9 \times 10^9 \text{ N.m}^2/\text{C}^2$

Coulomb's Law (contd.)

- Two balloons are charged with an identical quantity and type of charge: -6.25 nC. They are held apart at a separation distance of 61.7 cm. Determine the magnitude of the electrical force of repulsion between them.

Given:

$$Q_1 = -6.25 \text{ nC} = -6.25 \times 10^{-9} \text{ C}$$

$$Q_2 = -6.25 \text{ nC} = -6.25 \times 10^{-9} \text{ C}$$

$$d = 61.7 \text{ cm} = 0.617 \text{ m}$$

Find:

$$F_{\text{elect}} = ???$$

The final step of the strategy involves substituting known values into the Coulomb's law equation and using proper algebraic steps to solve for the unknown information. This substitution and algebra is shown below.

$$F_{\text{elect}} = k \cdot Q_1 \cdot Q_2 / d^2$$
$$F_{\text{elect}} = (9.0 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2) \cdot (6.25 \times 10^{-9} \text{ C}) \cdot (6.25 \times 10^{-9} \text{ C}) / (0.617 \text{ m})^2$$

$$F_{\text{elect}} = 9.23 \times 10^{-7} \text{ N}$$

Coulomb's Law (contd.)

- Two balloons with charges of $+3.37 \mu\text{C}$ and $-8.21 \mu\text{C}$ attract each other with a force of 0.0626 Newton . Determine the separation distance between the two balloons.

Given:

$$Q_1 = +3.37 \mu\text{C} = +3.37 \times 10^{-6} \text{ C}$$

$$Q_2 = -8.21 \mu\text{C} = -8.21 \times 10^{-6} \text{ C}$$

$$F_{\text{elect}} = -0.0626 \text{ N (use a - force value since it is attractive)}$$

Find:

$$d = ???$$

$$F_{\text{elect}} = k \cdot Q_1 \cdot Q_2 / d^2$$

$$d^2 \cdot F_{\text{elect}} = k \cdot Q_1 \cdot Q_2$$

$$d^2 = k \cdot Q_1 \cdot Q_2 / F_{\text{elect}}$$

$$d = \text{SQRT}(k \cdot Q_1 \cdot Q_2 / F_{\text{elect}})$$

$$d = \text{SQRT} [(9.0 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2) \cdot (-8.21 \times 10^{-6} \text{ C}) \cdot (+3.37 \times 10^{-6} \text{ C}) / (-0.0626 \text{ N})]$$

$$d = \text{Sqrt} [+3.98 \text{ m}^2]$$

$$d = +1.99 \text{ m}$$

Coulomb's Law (contd.)

- Two point charged particles are 4.41cm apart. They are moved and placed in a new position. The force between them is found to have been tripled. How far apart are they now?

Solution: initial distance is $r_1 = 4.41$ cm. At the new location, the force is tripled $F_2 = 3F_1$. Applying Coulomb's law, we have

$$F_2 = 3F_1$$

$$k \frac{|qq'|}{r_2^2} = 3k \frac{|qq'|}{r_1^2}$$

$$\frac{1}{r_2^2} = \frac{3}{(4.41 \times 10^{-2})^2}$$

$$\Rightarrow r_2^2 = \frac{(4.41 \times 10^{-2})^2}{3}$$

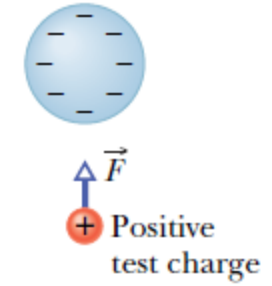
Taking the square root of both sides, we get

$$r_2 = 0.0254 \text{ m}$$

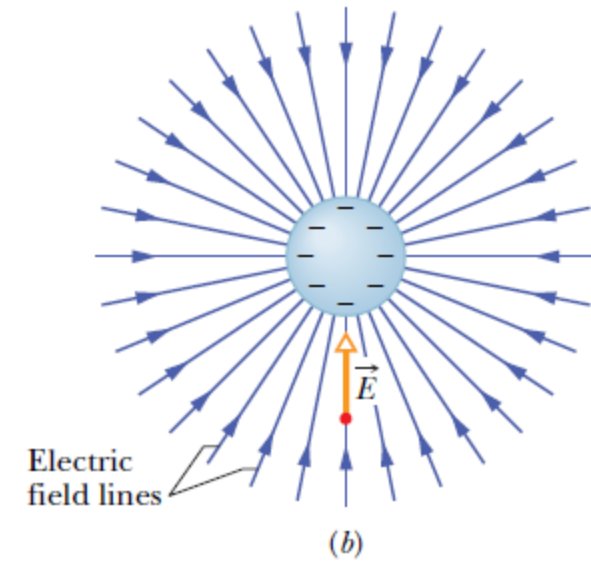
Thus, if those two charges are 2.54 cm away, the electrostatic force between them gets tripled.

Electric Field and Field Lines

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(a)



(b)

Electric field due to a point charge

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$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad (\text{point charge}).$$

We can quickly find the net, or resultant, electric field due to more than one point charge. If we place a positive test charge q_0 near n point charges $q_1, q_2, \dots, q_n$, then, the net force $\vec{F}_0$ from the n point charges acting on the test charge is

$$\vec{F}_0 = \vec{F}_{01} + \vec{F}_{02} + \dots + \vec{F}_{0n}.$$

the net electric field at the position of the test charge is

$$\begin{aligned} \vec{E} &= \frac{\vec{F}_0}{q_0} = \frac{\vec{F}_{01}}{q_0} + \frac{\vec{F}_{02}}{q_0} + \dots + \frac{\vec{F}_{0n}}{q_0} \\ &= \vec{E}_1 + \vec{E}_2 + \dots + \vec{E}_n. \end{aligned}$$

Mathematical Problem

- The nucleus of a plutonium-239 atom contains 94 protons. Assume that the nucleus is a sphere with radius 6.64 fm and with the charge of the protons uniformly spread through the sphere. At the nucleus surface, what are the (a) magnitude and (b) direction (radially inward or outward) of the electric field produced by the protons?
- **Solution**

(a) The magnitude of the total charge is Ze , so

$$E = \frac{Ze}{4\pi\epsilon_0 R^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(94)(1.60 \times 10^{-19} \text{ C})}{(6.64 \times 10^{-15} \text{ m})^2} = 3.07 \times 10^{21} \text{ N/C}.$$

(b) The field is normal to the surface and since the charge is positive, it points outward from the surface.

Mathematical Problem

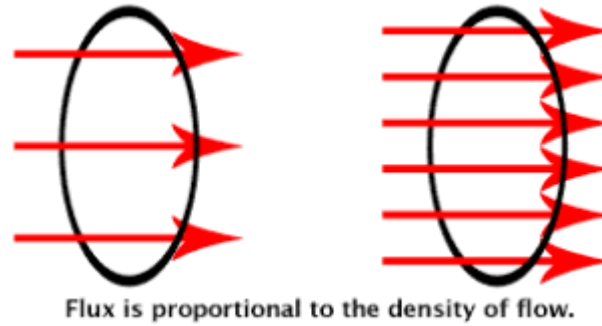
- What is the magnitude of a point charge whose electric field 50 cm away has the magnitude 2.0 N/C?
- **Solution:**

Since the magnitude of the electric field produced by a point charge q is given by $E = |q| / 4\pi\epsilon_0 r^2$, where r is the distance from the charge to the point where the field has magnitude E , the magnitude of the charge is

$$|q| = 4\pi\epsilon_0 r^2 E = \frac{(0.50 \text{ m})^2 (2.0 \text{ N/C})}{8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2} = 5.6 \times 10^{-11} \text{ C}.$$

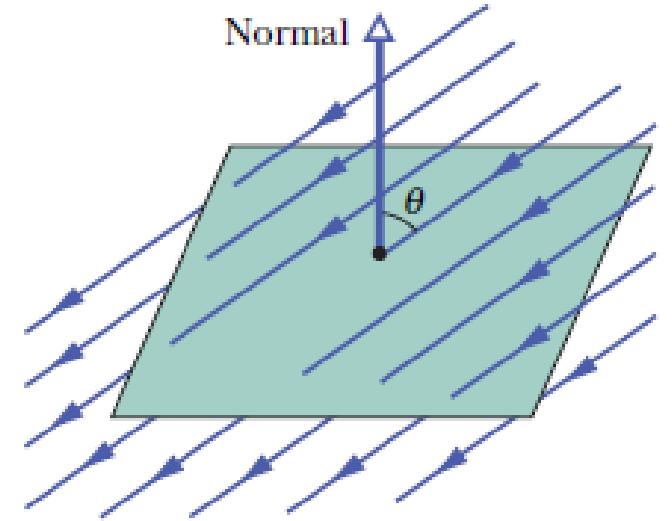
Flux of Electric Field

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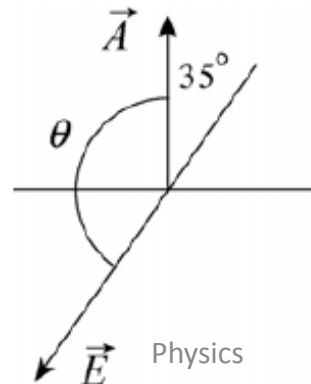
Mathematical Problem

- The square surface shown in figure measures 3.2 mm on each side. It is immersed in a uniform electric field with magnitude $E = 1800 \text{ N/C}$ and with field lines at an angle of 35° with a normal to the surface, as shown. Calculate the electric flux through the surface.
- Solution**



The vector area $\vec{A}$ and the electric field $\vec{E}$ are shown on the diagram below. The angle θ between them is $180^\circ - 35^\circ = 145^\circ$, so the electric flux through the area is

$$\Phi = \vec{E} \cdot \vec{A} = EA \cos \theta = (1800 \text{ N/C})(3.2 \times 10^{-3} \text{ m})^2 \cos 145^\circ = -1.5 \times 10^{-2} \text{ N} \cdot \text{m}^2/\text{C}.$$

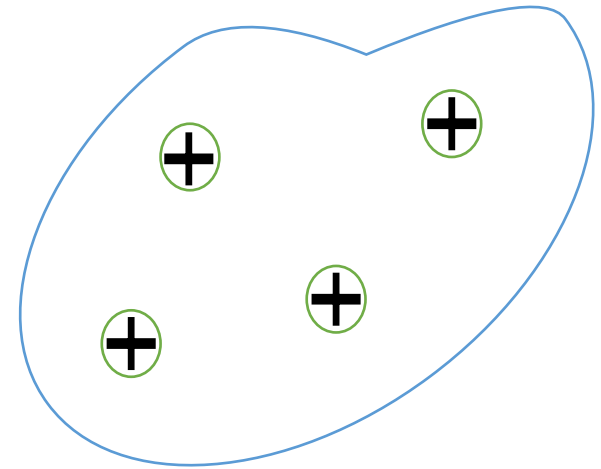


Practice Problem

- A solid ball with 0.5 meters radius has $10\text{ }\mu\text{C}$ electric charge in its center. Determine the electrical flux pass through the solid ball.
- Solution: Solve it as your practice and share the solution on google classroom please.
- Hints: Find Electric Field, Surface area and then electric flux

Gauss's Law

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Gauss's Law (contd.)

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Problems

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